Edge-to-Cloud Cyber-Physical System for Real-Time Pothole Detection, Multi-Object Tracking, and Geospatial Work-Order Remediation

Arush [Author Name], Student Member, IEEE | **[Faculty Advisor / Co-Author]**, Senior Member, IEEE
Department of Computer Science & Engineering • [Institute of Technology / University]
Email: arush@institution.edu, advisor@institution.edu

Abstract*—Roadway surface distress, particularly potholes, accounts for thousands of vehicular fatalities, severe chassis structural damage, and billions in municipal liability annually. Conventional road surveys rely either on hazardous manual walking audits or reactive smartphone accelerometer vibration crowdsensing, both of which fail to gauge visual distress geometry or generate reliable municipal work orders. In this paper, we propose a cyber-physical edge-to-cloud architecture for automated real-time road hazard perception, stateful multi-object tracking, and geospatial work-order remediation. The perception edge leverages an anchor-free single-stage YOLOv8 deep neural network, achieving a latency of 1.1 ms on an NVIDIA RTX 4060 GPU and 28.4 ms on CPU. To prevent telemetry flood caused by continuous video observation, we introduce temporal multi-object tracking via ByteTrack with an 8-state Kalman Filter and Kuhn-Munkres bipartite matching, cutting municipal bandwidth consumption by over 95%. Geolocation drift across multi-pass fleet patrols is reconciled using a 5-meter spherical Haversine spatial clustering engine on the backend. Detected anomalies are triaged into three severity tiers (Low, Medium, High) based on normalized surface bounding-box area ratios and routed through a dual-engine (MongoDB/In-Memory) RESTful backend into an interactive municipal Leaflet.js GIS command portal. Extensive empirical evaluations across diverse weather and lighting conditions demonstrate an mAP@0.5 of 94.2%, a precision of 91.8%, and a recall of 92.6%, verifying that the system satisfies hard real-time vehicular safety constraints.

***Index Terms*—Object Detection, YOLOv8, Multi-Object Tracking, ByteTrack, Edge Computing, Haversine Geospatial Clustering, Smart Cities, Intelligent Transportation Systems.**

I. INTRODUCTION

Road transport networks represent the most vital economic arteries of municipal commerce. However, environmental weathering, cyclic moisture infiltration, and heavy axle fatigue induce localized structural failure in asphalt pavements, culminating in potholes, alligator cracking, and surface depressions. Unmonitored pavement degradation directly threatens motorist safety, contributes to thousands of two-wheeler vehicular crashes annually, and imposes substantial municipal maintenance liabilities.

Historically, municipal Pavement Management Systems (PMS) have relied on two primary assessment paradigms: 1) *Manual Visual Condition Surveys*, which are labor-intensive, hazardous, and cover less than 5% of a municipal road grid per quarter; and 2) *Inertial Crowdsensing* via smartphone accelerometers. While inexpensive, vibration sensing is inherently reactive—the vehicle must physically hit the crater before detection occurs—and suffers from intolerable false-positive rates triggered by expansion joints, speed humps, and drainage grates.

Recent advances in Computer Vision (CV) have demonstrated the feasibility of optical road surveys. Nevertheless, existing academic literature predominantly treats pothole detection as an isolated, offline image classification task on static benchmark datasets, neglecting the cyber-physical realities of vehicular deployment: high vehicle velocity, continuous video frame duplication, cellular network latency, and geospatial database pin explosion.

A. Key Research Contributions

- **Sub-2ms Edge Perception:** We deploy an anchor-free YOLOv8 convolutional architecture optimized for non-rigid road defect geometry, achieving 1.1 ms GPU inference.
- **Stateful Multi-Object Tracking:** We incorporate ByteTrack with an 8-state Kalman Filter and Hungarian bipartite matching, eliminating duplicate network transmissions by 96.7%.
- **Spherical Haversine Clustering:** We formulate a 5-meter great-circle spatial deduplication engine on the cloud tier that reconciles GPS drift across patrol vehicle passes.
- **Dual-Image Zoom Evidence:** We implement an automated severity triage metric paired with high-resolution 25% padded defect cropping for municipal engineering audit compliance.

II. RELATED WORK

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Early road hazard vision systems relied on classical image processing pipelines, including Canny/Sobel edge filters, Otsu thresholding, and Gabor wavelets paired with Support Vector Machines (SVM). These methods fundamentally failed under real-world conditions due to high intra-class variance and sensitivity to tree shadows and puddle specular reflections.

With the emergence of deep Convolutional Neural Networks (CNNs), two-stage detectors such as Faster R-CNN achieved high localization accuracy via Region Proposal Networks (RPN). However, their computational complexity limits throughput to 10-14 FPS, violating vehicular real-time constraints. Single-stage anchor-based detectors (e.g. YOLOv5, SSD) rely on rigid predefined anchor boxes, which struggle to fit the organic aspect-ratio variance of asphalt craters.

III. PROPOSED SYSTEM ARCHITECTURE

The proposed cyber-physical system is structured into three decoupled tiers: the *Perception Edge Tier*, the *Telemetry Ingestion Middleware Tier*, and the *Municipal GIS Command Tier*.

A. Edge Perception Pipeline

Let $I_t \in \blacksquare^{H \times W \times 3}$ denote the optical video frame captured at time t. The tensor undergoes Letterbox resizing and RGB normalization $X_t = \Gamma(I_t) \in \blacksquare^{640 \times 640 \times 3}$. The forward pass yields bounding boxes B_t and class confidence scores C_t:

$$B_t, C_t = f_\theta(X_t)$$

Spatial Non-Maximum Suppression (NMS) suppresses overlapping candidates exceeding IoU threshold $\tau = 0.45$.

B. Complete IoU (CIoU) Loss Formulation

YOLOv8 optimizes Complete IoU loss, penalizing overlap, centroid distance, and aspect ratio consistency simultaneously:

$$L_{CIoU} = 1 - IoU + (\rho^2(b, b_{gt}) / c^2) + \alpha\nu$$

where $\nu = (4/\pi^2)(\arctan(w_{gt}/h_{gt}) - \arctan(w/h))^2$ and $\alpha = \nu / ((1 - IoU) + \nu)$.

C. Temporal Multi-Object Tracking (ByteTrack)

As a vehicle drives past a pothole at 60 km/h, the defect remains in camera field-of-view for 15-30 frames. Unchecked, this produces 30 duplicate database records. We model defect motion using an 8-state continuous Kalman Filter:

$$x = [u, v, a, h, u\blacksquare, v\blacksquare, a\blacksquare, h\blacksquare]^T$$

Tracks and detections are associated by minimizing spatial cost $C_{i,j} = 1 - IoU(T_i, D_j)$ via the Hungarian algorithm. When a track is confirmed, track ID k is registered in an in-memory set. Subsequent frames update the visual bounding box on screen but completely suppress duplicate network payloads.

D. Severity Triage Metric

Severity S is computed as the normalized ratio of defect bounding box area to total frame area:

$$R = (w_{bbox} \times h_{bbox}) / (W_{frame} \times H_{frame})$$

Defects are categorized into Low ($R < 0.02$), Medium ($0.02 \leq R \leq 0.06$), and High ($R > 0.06$), with High severity triggering immediate non-blocking audio alerts.

E. Spherical Haversine Spatial Clustering

Consumer vehicular GPS chips exhibit 2-4m horizontal circular error probable (CEP). When subsequent patrol vehicles re-observe a defect, the backend computes great-circle distance:

$$a = \sin^2(\Delta\phi/2) + \cos(\phi1)\cos(\phi2)\sin^2(\Delta\lambda/2)$$
$$d = 2\, R_{earth} \times \operatorname{atan2}(\sqrt{a}, \sqrt{(1 - a)})$$

If $d \leq 5.0$ *meters*, the detection is merged into the existing cluster, updating cluster frequency; otherwise, a new cluster entity is created.

IV. EXPERIMENTAL SETUP & BENCHMARKS

The model was trained on 3,850 road distress images from the RDD2022 corpus and regional road surveys. Augmentations included Mosaic 4-image stitching, affine warping, and synthetic rain/glare simulation.

Training executed on an NVIDIA RTX 4060 GPU (8GB VRAM) under PyTorch 2.5 and CUDA 12.4. Hyperparameters: Batch size 16, SGD optimizer (momentum 0.937, weight decay 0.0005), 50 epochs with Cosine Annealing learning rate decay (0.01 to 0.0001).

Model Architecture	mAP@0.5	Precision	Recall	Latency
Faster R-CNN (ResNet-50)	88.4%	85.1%	86.9%	84.2 ms
SSD (MobileNet-V2)	81.2%	79.4%	78.6%	14.5 ms
YOLOv5s (Anchor-based)	89.6%	87.3%	88.0%	4.8 ms
YOLOv8n (Proposed)	94.2%	91.8%	92.6%	1.1 ms

V. RESULTS & DISCUSSION

As reported in Table I, our proposed YOLOv8n pipeline delivers superior detection accuracy (94.2% mAP@0.5) while running at a staggering 1.1 ms forward pass latency on the GPU. On CPU, inference completes in 28.4 ms, sustaining >30 FPS real-time operation.

A. Deduplication Ablation Analysis

In a live 5 km road test featuring 14 physical potholes, the baseline detector without tracking transmitted 428 HTTP alerts. With ByteTrack active, alerts dropped to exactly 14—a 96.7% reduction in network overhead and database write contention.

VI. CONCLUSION & FUTURE WORK

We presented an end-to-end cyber-physical road inspection system uniting anchor-free YOLOv8 vision, ByteTrack temporal deduplication, and 5-meter Haversine spatial clustering. In future work, we plan to integrate stereo-vision disparity matching for volumetric asphalt calculations (m^3) and automated drone repair validation.

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